

7b local re-audit and breakdown status — 2026-09-27 UTC

Scope and outcome

Owner requested another audit and an assessment of progress through the breakdown. This pass inspected current adjacent source diffs, repair propagation, test preservation, report inventories, GitHub review bodies, and downloaded results from E's final CI. It did not request another Copilot review, create another split, rebuild firmware, or perform a physical-device test. The owner's instruction to preserve the existing drafts remains in force.

No new firmware defect was confirmed in the paths inspected. One new reporting defect was confirmed below. This is a bounded re-audit, not proof that every security property or every historical finding has been independently closed.

Current breakdown

Unit	Firmware draft / immutable head	Review attempts	Status
Original complete 7b	#863 / 729ae7cfacf581ce534a98f6f5e0cc39efee3838	3; seven findings repaired	Preserved historical candidate; replaced for review by the following three units.
A–C: formatters and interpolated intent	#865 / 5c0090e038769ed36d627d4dca09cbfcb1df91d3	1	Findings: None, body observations dispositioned in comment 5849910641; current-head CI passed.
D: groups, iteration, optional visibility	#866 / 26ae09fe30e38714d49ffa9aadb34010fed2b8ab	3	Four inline findings repaired/resolved; current-head CI passed; no clean review of the repaired head.
E: embedded calls	#867 / 19d7a30251bf62c931d7b058788a3939edad01da	0	Includes A–C and D; current-head CI passed; external review incomplete.

One of three replacement review units has passed its external checkpoint. All three units are implemented in the E stack. This is a count of review units, not a percentage of engineering work or release readiness. Seven Copilot reviews have been used across historical #863 and its replacements: three original, one A–C, three D, zero E.

The proposed next subdivision is D groups/visibility followed by D iteration, with E replayed on that accepted base. Those replacement units have not been created. The code is already stacked together in E; accepted recombination still requires the remaining review and integration checkpoints.

The larger release effort is currently at 7b. Stack 08's owner receipt records provisional embedded-call tests, not acceptance. Live downstream PRs remain PR #838 (08), #839 (09), #840 (10), #842 (12), and #843 (13). This pass did not re-audit those blocks or infer completion from their existence. Earlier master ledger entries also retain predecessor-integration and owner-exception limits.

Source and preservation checks

Report inventories match the actual adjacent diffs exactly: A–C 31 paths, D 14, E 25. D/E worktrees are clean. A–C has an untracked `scripts/emulator/__pycache__` directory; no tracked modification was found.

Compared with historical #863, the assembled E production source differs in only three files: `erc7730_catalog.h`, `erc7730_catalog.c`, and `fsm_msg_ethereum.h`. Those changes are D's strict formatter-array binding, the associated formatter flag allocation, and guarded capture-count decoding. The inspected E display verifier uses the updated flag bits consistently.

Reviewed D's active-array ownership, scalar/auxiliary path checks, separate display frames, bounded count capture, numbered field continuation, and the fixed-array wire assertions. Reviewed E's inner definition chain/callee/selector binding, byte-identical outer definition restoration, missing-value refusal, and the restriction that embedded calls inside iteration are shown blind. The runtime test source covers wrong inner bindings, replay mutation, missing amount, depth-two fallback, cancellation, and recovery after a refused inner call.

E removes 17 test methods from the local Stack 07 script. Sixteen retain their names in the pinned Python runtime module. The remaining native-alias test was renamed to `test_native_alias_discloses_signer_mapping_and_token_address` and explicitly asserts signer provenance, alias address, and amount. No method is unaccounted for after this reviewed rename. E's changed

native test files retain all D TEST/TEST_F declarations and add four. Declaration preservation alone is not behavioral equivalence; the runtime assertions and CI results provide the additional evidence described here.

Executed evidence inspected

Downloaded artifacts from [E CI 36286655387](#), which completed successfully on 19d7a30251bf62c931d7b058788a3939edad01da.

Artifact suite	Passed	Skipped	Failures/errors
Full native firmware	656	0	0
Bitcoin-only native firmware	135	0	0
Full Python main suite	791	89	0
Bitcoin-only Python main suite	341	539	0
Full Stack 07 wire suite	21	0	0
Bitcoin-only Stack 07 wire suite	1	20	0
Full screenshot suite	189	26	0
Bitcoin-only screenshot suite	40	175	0

All 38 ERC-7730 runtime methods passed in the full Python suite; 28 also ran in the screenshot suite. All 38 deliberately skip in bitcoin-only. The fixed and dynamic array cases are assertions inside one method, not separate test counts. The native, Python, screenshot, and merged report rows overlap and must not be summed as independent tests.

The generated report manifest identifies the correct firmware head and Python pin `fb94c5c2cd6cb5e3b0b3d896a7c58fc0a8ad0a42`. Its PDF SHA256 matches the downloaded PDF. Full and bitcoin-only ARM, static analysis, report generation, CI gate, and release-evidence gate completed successfully. Existing CI is being inspected here; no new build or test run is claimed.

Reproducible inventory/test/artifact checks and hashes are retained in the sibling `715-07b-reaudit-20260927-checks.json`. Downloaded raw artifacts are under `/private/tmp/kk715-reaudit-20260927`; their authoritative source is the immutable CI run above.

Findings and remaining evidence gaps

ID	Finding	Disposition
RA-01	E's generated manifest sets <code>python_pr</code> to <code>keepkey/python-keepkey#197</code> , while the current companion is <code>BitHighlander/python-keepkey#116</code> . <code>.github/workflows/ci.yml:1035</code> hardcodes the old link. The Python SHA itself is correct.	Open reporting defect. Correct the workflow's companion provenance when preparing the next candidate; do not relabel the already generated artifact as corrected.
RA-02	Frozen D/E source reports and PDFs retain preparation-time pending CI text; the live PR descriptions and master handoff record the successful final runs.	Reconciled by this dated addendum. The frozen reports are historical snapshots, not a standalone current-status source.
RA-03	Original 42 raw audit rows were not retained; D13/D19 are unnamed in the surviving cluster receipt.	Historical evidence gap remains open. Do not infer individual closure from cluster counts or green tests.
RA-04	Split reports record each review's repairs and checks, but do not establish a full independent audit and retrospective after every D round.	This retrospective adds analysis now; it cannot prove that missing historical work occurred before each request.

The inherited canonical preflight exceptions remain recorded: duplicate `nanopb` options, a missing predecessor report-gate test, and the script's missing `cppcheck` build directory. The prior push exception does not make that gate passing. No physical-device, merge, or release acceptance is established.

Retrospective of D's three rounds

Round	What actually failed	Why the earlier check was insufficient	Prevention for the next candidate
1	Auxiliary formatter paths could escape the active array; failed capture could read uninitialized count bytes.	Checking only the value path omitted the other formatter inputs; successful-capture tests did not exercise error-path reads.	Trace every formatter argument to its active array and audit all capture failure exits before UI continuation; retain same-array, different-array and scalar controls.
2	Handoff still said 1,294 signable formats after strict array binding reduced D to 1,273.	The code/test correction was not propagated to every statement of the result. This was a documentation defect, not a new firmware defect.	Search all handoffs and report summaries after a behavior/count change; bind current-status receipts to the exact tested head.
3	Dynamic/empty arrays had wire coverage; fixed ABI arrays did not.	Similar path handling was treated as sufficient despite fixed arrays having no encoded length word.	Require a matrix of dynamic, fixed, empty, short, tuple and malformed cases at the actual wire entry point.

All four inline findings are fixed and resolved. A repaired finding and green CI do not constitute a fresh clean Copilot review. The three-attempt budget is exhausted; no fourth request or further split was performed in this pass.

The re-audit's own lesson is to inspect the generated artifact, not just its successful job: that inspection exposed RA-01. Future acceptance should require the firmware head, Python pin, companion PR link, and final test inventory to agree in one current receipt. Review convergence and release acceptance remain separate from passing tests.